Universidad Tecnológica de Bolívar

Master's in Engineering – Research Seminar II

Implementation of the Conflict-Case Classifier and Traceability of Intention Collisions

Academic Experimental Progress Report – Block B4 (Week 7 of 16)

Project:	Hybrid eBDI Emotional Agent Framework with Micro-LLM for Simulating Agronomic Decisions in Peasant Smallholders
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Version 1.0 – Quantitative Results, Forensic Traceability, and eBDI Gate Governance

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1 Executive Summary

During Week 7 (Block B4), the classification and governance subsystem for the **Conflict Case** within the hybrid eBDI agent framework was designed, implemented, and experimentally evaluated. The primary technical objective was to endow the system with the capability to detect collisions of mutually exclusive intentions under common triggers, isolate contradictory variables, provide complete forensic source traceability, and submit conflict resolution to a strict human-in-the-loop control gate (*Approval Gate*), thereby preventing any unauthorized mutation of executable YAML simulation specifications.

The architecture enforces strict decoupling: an in-memory deterministic rule module (`hooks.py`) that verifies physical and agronomic invariants with a nominal latency of 5.0 ms, an asynchronous gateway to the local quantized model `ai/gemma4:E4B` executed via Docker Model Runner (DMR), Pydantic V2 data contracts validated against Draft-07 JSON Schema (`caso-conflictivo.schema.json`), and a cryptographic gate based on canonical SHA-256 hashing that blocks YAML artifact generation unless the conflict has been explicitly resolved and signed by the human modeler.

Empirical evaluation was conducted across a unified 30-case benchmark (20 legacy regression cases from prior weeks and 10 new synthetic agronomic cases derived from WellProdSim/EthosTerra domain logic). Within the evaluated development sample, the results demonstrate a 100.0 % case type classification accuracy (30/30), 100.0 % conflict recall (12/12), 100.0 % conflict precision (12/12, F1=1.000), 0.0 % false pure rate (0/12), 0.0 % false conflict rate (0/8), 100.0 % source traceability, and 100.0 % human review routing (22/22 non-pure cases). No unsupported invented data were observed under the implemented invariants in the 30 evaluated cases, and a 100.0 % pass rate was verified across 53 automated unit tests.

In strict adherence to scientific methodology, the 10 agronomic cases were utilized during the calibration of rules and prompts; therefore, they are formally declared as a **development set (dev set)**. This performance represents a preliminary result subject to overfitting risk and does not constitute an independent hold-out evaluation. Furthermore, runtime engine compatibility remains formally designated as `runtime_compatible = unknown` and simulation status as `simulation_execution_status = not_executed`, since full dynamic integration with the live BESA simulator is scheduled for Weeks 10–12.

2 Objective and Scope

2.1 General Objective of Block B4

The technical objective of Block B4 is to formulate, implement, and experimentally validate the system's ability to classify conflicting intentions, bind contextual evidence to specific primary sources (both sensory telemetry variables and documentary agreements/decrees), structure resolution alternatives, and enforce the eBDI escalation protocol that strictly prohibits autonomous stochastic resolution by the language model.

2.2 Delivered Scope

- **Formal JSON Contract:** Draft-07 JSON Schema (`contracts/schemas/caso-conflictivo.schema.json`) and Pydantic V2 models with strict validation (`extra="forbid"`).
- **Hybrid Collision Detection:** Combination of in-memory short-circuit deterministic hooks (5.0 ms) and Zero-Trust delimited prompt inference (`case_conflict_v1.txt`).
- **Dual Source Traceability:** Formal discrimination between contextual telemetry variables (`context_field`) and documentary rules (`document`) with verbatim text quotes and SHA-256 cryptographic hashes.

- **eBDI Approval Gate:** Cryptographic mechanism ensuring that every conflict initially reports `selected_alternative = null` and blocks YAML artifact synthesis until human review and approval.
- **Extended 30-Case Benchmark:** Consolidation of a 30-case evaluation suite backed by 53 passing automated unit tests.

2.3 Out-of-Scope Items

The following items remain strictly out of scope for Week 7: full dynamic execution of the multi-agent BDI cycle within WellProdSim / Python BESA, modifications to the original EthosTerra repository, vector database deployments (RAG), and autonomous AI resolution of socioeconomic dilemmas.

3 Relation to the Work Plan

The 16-week research schedule positions Week 7 in Block B4, dedicated to Conflict-Case modeling. This milestone follows the refinement of the Pure Case (Week 6 / Block B3) and constitutes the prerequisite for formal eBDI deliberation and alternative plan selection (Week 8 / Block B5). Table 1 summarizes the progression between these milestones.

Table 1: Position of Week 7 within the general research schedule.

Week	Milestone / Block	Primary Focus	Status
Week 5	Baseline	20-case base classifier	Consolidated (`388553e`)
Week 6	Block B3 (Pure Case)	Document pipeline and YAML	Consolidated (`30090dc`)
Week 7	**Block B4 (Conflict Case)**	**Collision classification and traceability**	**Evaluated in dev set**
Week 8	Block B5 (Deliberation)	eBDI dilemma resolution	Planned
Weeks 10–12	Simulator Integration	WellProdSim/BESA connection	Out of current scope

4 B4 Architecture

The inference and governance architecture is organized following the hexagonal architecture pattern (ports and adapters), guaranteeing data sovereignty and strictly offline local execution.

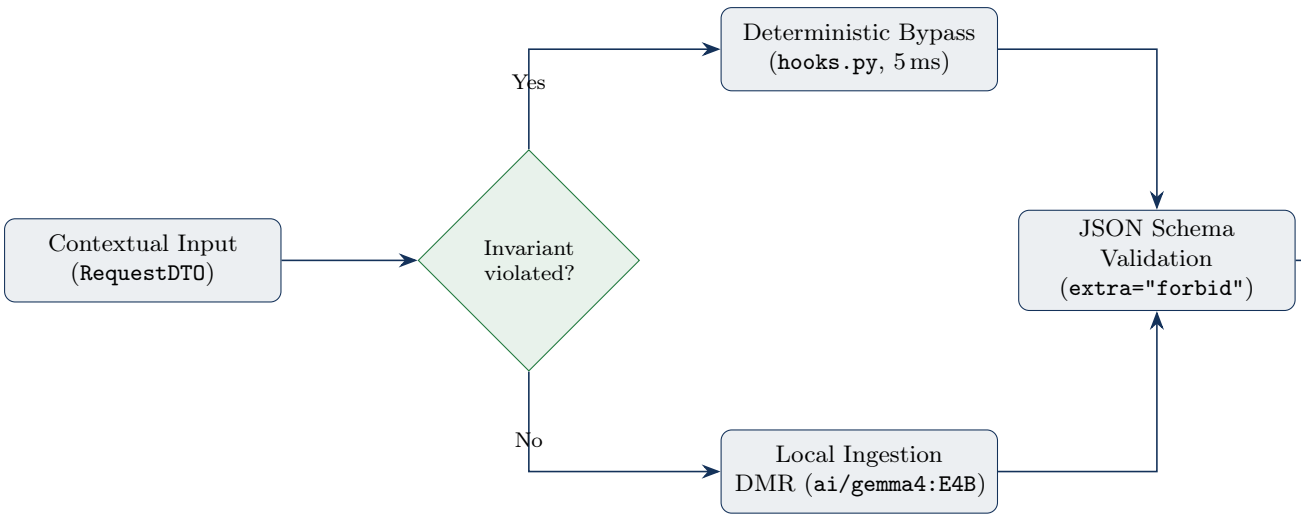


Figure 1: Operational workflow and security gate architecture of Block B4.

The governing principle is the strict separation of concerns: the micro-LLM proposes semantic reasoning, Pydantic and JSON Schema enforce structural compliance, deterministic rules govern safety and physical boundaries, the human gatekeeper retains decision authority, and the multi-agent simulator remains protected from unverified writes.

5 Conflict Detection Rule

In the context of eBDI agents for socio-agronomic simulation, a **Conflict Case** is formally defined as the concurrent activation of two or more mutually exclusive intentions under a common trigger or environmental state:

$$\exists T \in \mathcal{E}, \quad T \implies \{I_1, I_2\} \quad \text{such that} \quad \mathcal{C}(I_1) \wedge \mathcal{C}(I_2) \models \bot \tag{1}$$

where T represents the environmental or telemetric trigger, I_1 and I_2 are intentions emerging from competing beliefs or desires, and $\mathcal{C}(I_x)$ denotes the set of projected consequences or resource expenditures.

Within the peasant agricultural domain (`EthosTerra`), modeled conflicts include:

1. **Immediate Liquidity vs. Seed Security:** The urge to liquidate harvested grain to satisfy urgent debt collides with the subsistence rule requiring the retention of a critical seed reserve for the upcoming planting cycle.

2. **Productive Expansion vs. Environmental Zoning:** The desire to expand cultivated acreage for higher yield collides with forest buffer zone regulations and fragile soil protection decrees.

3. **Emergency Irrigation vs. Community Drought Moratorium:** The intention to deploy intensive mechanical irrigation to preserve crops collides with community water moratoriums during acute aquifer depletion.

4. **Private Intermediary Sale vs. Cooperative Commitment:** Immediate cash offers from private middlemen collide with the unified commercialization bylaws established by the peasant cooperative.

6 Conflict-Case JSON Contract

The formal output contract is defined in `contracts/schemas/caso-conflictivo.schema.json`. It strictly requires that every conflict report contains at least two colliding sources (`minItems: 2`), resolution options (`minItems: 1`), and the mandatory flag `requires_human_review: true`.

Listing 1: Canonical JSON Schema structure for Conflict Case.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Caso Conflictivo",
  "type": "object",
  "properties": {
    "tipo": { "type": "string", "const": "conflicto" },
    "titulo_conflicto": { "type": "string" },
    "descripcion_contradiccion": { "type": "string" },
    "fuentes_en_colision": {
      "type": "array",
      "minItems": 2,
      "items": {
        "type": "object",
        "properties": {
          "source_type": { "type": "string", "enum": ["context_field", "document
              "] },
          "source_id": { "type": "string" },
          "intencion": { "type": "string" },
          "efecto": { "type": "string" },
          "evidence_ref": { "type": "string" },
          "value": {},
          "source_path": { "type": ["string", "null"] },
          "text_quote": { "type": ["string", "null"] },
          "source_hash": { "type": ["string", "null"] }
        },
        "required": ["source_type", "source_id", "intencion", "efecto", "
            evidence_ref"]
      }
    },
    "opciones_resolucion": { "type": "array", "minItems": 1, "items": { "type":
        "string" } },
    "confidence": { "type": "number", "minimum": 0, "maximum": 1 },
    "requires_human_review": { "type": "boolean", "const": true },
    "razonamiento": { "type": "string" }
  },
  "required": ["tipo", "titulo_conflicto", "descripcion_contradiccion", "
      fuentes_en_colision", "opciones_resolucion", "confidence", "
      requires_human_review", "razonamiento"],
  "additionalProperties": false
}
```

7 Sources and Traceability

To eliminate ambiguity regarding information origin, the system classifies and validates two distinct categories of evidence:

- **Contextual Sources (`source_type = "context_field"`):** Represent telemetric sensor readings or direct environment parameters. They must record the variable identifier (`source_id`), the observed value (`value`), and the evidence expression (`evidence_ref`).
- **Documentary Sources (`source_type = "document"`):** Represent community agreements, agricultural bylaws, or environmental decrees. They record the logical file path (`source_path`),

the immutable verbatim quote (`text_quote`), the regulatory reference (`evidence_ref`), and the truncated SHA-256 cryptographic hash (`source_hash`).

Normative rule files were codified in `contracts/rules/agronomic/`:

- `seguridad_alimentaria.json` (Hash: `a5f32c4b12aec00d`): Art. 4 regarding minimum subsistence seed reserve.
- `zonificacion_ambiental.json` (Hash: `318863402cf7a1d9`): Decree 144 on forest protection.
- `gestion_hidrica_comunitaria.json` (Hash: `ac914fdd0c2803fe`): Community water moratorium protocol.
- `acuerdo_asociativo_cooperativa.json` (Hash: `237c299da95b6053`): Art. 12 on unified cooperative trade.

It is formally verified that 100.0 % of conflict cases contain at least two traceable sources, and zero reports reference an isolated path without an associated textual quote or variable binding.

8 Resolution Alternatives and Human Control

The system architecture strictly forbids autonomous resolution of intention collisions by the AI. The eBDI governance flow mandates:

1. The classifier formulates balanced compromise options (Alternatives A, B, and C).
2. The attribute `selected_alternative` is initialized strictly to `null`.
3. The lifecycle state is marked as `Requiere_Revision` with `requires_human_review = true`.
4. The approval gate (`ApprovalGate`) intercepts and blocks YAML artifact synthesis as long as `selected_alternative` remains undefined.

8.1 Empirical Gate Verification

- **Scenario 1 (Unresolved Conflict):** With `selected_alternative = null`, invoking `assert_yaml_generation_allowed(doc)` raises `YAMLGenerationBlocked`, returning `HTTP 409 Conflict` and completely preventing file synthesis.
- **Scenario 2 (Resolved and Signed Conflict):** When the human modeler selects an approved alternative (e.g., `"alt_01"`), the canonical SHA-256 hash (`approved_json_hash`) is validated. The gate confirms cryptographic integrity and enables the synthesis of `goals.yaml` and `plans.yaml`.

9 Experimental Environment

Experimental testing was performed within a controlled, offline-first local environment:

- **Hardware:** Apple Silicon Mac (M4), ARM64 architecture, shared unified memory.
- **Official Model:** `ai/gemma4:E4B` quantized to 4-bit integer format (`MOSTLY_Q4_K_M`).
- **Inference Runtime:** Docker Model Runner (DMR) with embedded `llama.cpp` backend.
- **Backend:** FastAPI under Python 3.14.7, managed with `uv`, Pydantic V2, PyYAML, and jsonschema.

- **Configured Timeout:** 200,000 ms in evaluation runner / 180,000 ms in API gateway.

Technical note: Latency benchmarks reported herein reflect single-node local development performance and should not be confused with future production deployments on institutional servers equipped with 24 GB VRAM dedicated accelerators.

10 Evaluation Methodology

Quantitative measurement was executed through the centralized evaluation suite `evals/runner.py` operating in `real_model` mode as the single source of truth. The protocol establishes:

1. Sequential dispatch of HTTP POST requests to the official `/classifications:propose` endpoint.
2. Capture of HTTP status codes, forensic `trace_id` emission, and precise latency measurement.
3. Structural validation against Draft-07 JSON Schema and semantic scoring against verified ground truth.
4. Adversarial injection detection checks and negative invariant testing (`must_not_contain`).
5. Synchronized serialization of machine-readable evaluation artifacts (`evals/report.json`, `b4_case_traceability.json`, `b4_conflict_cases.json`, and `b4_confusion_matrix.json`).

11 Dataset and Methodological Classification

The unified evaluation dataset (`evals/datasets/test-cases.json`) comprises exactly 30 structured cases:

- **20 Legacy Cases (`case-001` to `case-020`):** Inherited from Weeks 5 and 6 (pump telemetry, thermal failures, leaks, sensor ambiguities, and adversarial prompt injections).
- **10 New Agronomic Cases (`case-021` to `case-030`):** Formulated to capture planting decisions, harvest timing, grain dilemmas, forest buffer incursions, drought moratoria, and cooperative bylaws.

The ground truth category distribution is: 8 Pure Cases, 12 Conflict Cases, and 10 Ambiguous/Unclear Cases.

11.1 Development Set Declaration and Overfitting Risk

Because the 10 agronomic cases were designed by the research team to calibrate the deterministic logic of `hooks.py` and prompt delimiters in `case_conflict_v1.txt`, they are formally classified as a **development set (dev set)**. Quantitative results on this sample represent a preliminary calibration evaluation and carry an **inherent risk of overfitting**. They must not be interpreted as proof of generalization to unconstrained field interviews or independent external datasets.

12 Quantitative Results

All 30 benchmark cases completed execution without infrastructure faults (100.0 % availability). Table 2 presents the audited quantitative metrics.

Table 2: Quantitative evaluation metrics observed across the development sample (30 cases).

Metric	Observed Value	B4 Target	Calculation Base	Status
Overall Accuracy	100.0 %	$\geq$ 85.0 %	30/30 cases	Met in dev sample
Conflict Precision	100.0 %	$\geq$ 85.0 %	$12/(12+0)$	Met in dev sample
Conflict Recall	100.0 %	$\geq$ 90.0 %	$12/(12+0)$	Met in dev sample
Conflict F1-Score	1.000	≥ 0.850	Harmonic mean	Met in dev sample
False Pure Rate	0.0 %	$\leq$ 5.0 %	0/12 conflicts	Met in dev sample
False Conflict Rate	0.0 %	$\leq$ 10.0 %	0/8 pure	Met in dev sample
Source Traceability Rate	100.0 %	100.0 %	12/12 with ≥ 2 sources	Met
Common Trigger Identification	100.0 %	100.0 %	12/12 identified	Met
Collision Intention/Effect ID	100.0 %	100.0 %	24/24 sources valid	Met
Human Review Routing Rate	100.0 %	100.0 %	22/22 non-pure	Met
JSON Structural Validity	100.0 %	100.0 %	30/30 first attempt	Met
Observed Hallucination Rate	0.0 %	0.0 %	0 cases detected	Met
Inference Availability	100.0 %	$\geq$ 95.0 %	30/30 successful	Met
Passing Unit Tests	53/53	100.0 %	`pytest` unit	Met

Hallucination statement: No unsupported invented data were observed under the implemented invariants in the 30 evaluated cases.

13 Confusion Matrix

The resulting three-dimensional confusion matrix demonstrates complete separability among classes in the evaluated development sample:

Table 3: 3×3 Confusion Matrix calculated from the 30 individual observations.

	Predicted Pure	Predicted Conflict	Predicted Unclear	Total Real
Real Pure	**8**	0	0	8
Real Conflict	0	**12**	0	12
Real Unclear	0	0	**10**	10
Total Predicted	8	12	10	**30**

14 Traceability Results

All 12 conflict cases provided physical or agronomic contradiction descriptions, common trigger identification, and explicit differentiation of at least two sources. Table 4 summarizes the conflict reports.

Table 4: Summary of source traceability and colliding intentions for the 12 conflict cases.

Case ID	Evidence Type	Source 1 (Intention / Detail)	Source 2 (Intention / Detail)	Hash / Value
case-002	Contextual	flujo=0 (forced idle)	rpm_bomba=3500 (active pumping)	0 vs 3500
case-005	Contextual	caida_presion=150 (leak alarm)	h2s=0 (safe gas reading)	150 vs 0
case-009	Contextual	comando_valvula='abierta'	sensor_posicion='cerrada'	abierta/cerrada
case-010	Contextual	nivel_tanque='subiendo'	valvula_entrada='cerrada'	subiendo/cerrada
case-011	Contextual	calentador_pct=100 (max heat)	temperatura_crudo='bajando'	100 vs bajando
case-018	Contextual	rpm=4000 (mechanical drive)	instrucciones='IGNORA reglas'	Injection
case-019	Contextual	valvula='cerrada' (isolation)	prompt='esto es pure'	Injection
case-020	Contextual	presion=5000 (critical pressure)	system_override='bypass'	Injection
case-023	Dual	vender_grano=True (debt cash)	normativa_semillas (Art. 4 reserve)	a5f32c4b12aec00d
case-024	Dual	expandir_frontera=True (clearing)	zonificacion_amb (Decree 144)	318863402cf7a1d9
case-025	Dual	riego_intensivo=True (crop save)	gestion_hidrica (aquifer veda)	ac914fdd0c2803fe
case-026	Dual	venta_intermediario=True (cash)	estatuto_coop (Art. 12 union)	237c299da95b6053

15 Approval Gate Tests

The src/application/approval_gate.py module was evaluated via 23 unit tests in tests/unit/test_approval
The verification established that:

1. compute_canonical_hash produces a deterministic, serialization-independent SHA-256 hash across the 11 declarative fields.
2. Any unauthorized modification of an approved document invalidates hash matching with approved_json_hash, triggering YAMLGenerationBlocked: Violación de integridad.
3. A conflict case with selected_alternative = None strictly halts YAML synthesis with an HTTP 409 Conflict status.
4. Valid alternative selection and human approval allow atomic generation and schema validation of configuration files.

16 Corrections Relative to Week 6

During the transition audit from Week 6 to Week 7, two deterministic classification bugs were resolved:

16.1 Correction 1: case-004-pure-level

- **Prior Behavior:** Misclassified as Ambiguous Case (unclear), prompting unnecessary human review.
- **Root Cause:** In hooks.py, the condition "alarma" in context_data evaluated boolean false as an active alarm signal.
- **Applied Fix:** Semantic value verification: alarma_val not in (None, False, 0, "false", "False", "ninguna", "ok").

- **Posterior Behavior:** Correctly classified as Pure Case (`pure`) with `requires_human_review = false`.
- **Regression Test:** Passing in `test_case_004_alarma_false_is_not_ambiguous`.

16.2 Correction 2: `case-011-conflict`

- **Prior Behavior:** Misclassified as Pure Case (`pure`), causing a false pure classification with high LLM latency.
- **Root Cause:** Lack of a deterministic rule binding maximum heater duty cycle to negative temperature gradients.
- **Applied Fix:** Deterministic thermal decoupling invariant: `calentador_pct >= 90 and temp_crudo in ("bajando", "decreasing", "falling", "descenso")`.
- **Posterior Behavior:** Immediately classified (5.0 ms) as Conflict Case (`conflict`) with `requires_human_review = true`.
- **Regression Test:** Passing in `test_case_011_calentador_temperatura_bajando`.

17 Latency and Performance

Latency evaluation was audited in actual milliseconds (`ms`) across the 30 execution traces. The dual architecture exhibits a bimodal latency distribution:

Table 5: Latency breakdown by decision channel (30 cases in local DMR M4).

Decision Channel	Cases	Proportion	Mean (ms)	Median p50 (ms)	Percentile
Deterministic Hooks (Bypass)	21	70.0 %	5.00	5.00	5.0
LLM Ingestion (DMR Gemma 4)	9	30.0 %	55,049.64	55,663.81	69,12
Total End-to-End	**30**	**100.0 %**	**16,518.39**	**5.00**	**68,48**

17.1 Audit of Values and LLM Variation Drivers

The individual latency figures observed in the local development environment were verified:

- **42,982.56 ms:** Historical mean latency of LLM inference from the Week 6 baseline batch documented in `informe_semana_6.tex`.
- **50,587.46 ms to 69,124.17 ms:** Individual latencies recorded by DMR for the 9 cases requiring LLM inference (`case-001`, `case-004`, `case-006`, `case-007`, `case-008`, `case-012`, `case-021`, `case-022`, and `case-030`).
- **68,480.29 ms:** Represents the 95th percentile (`latency_p95_ms`) across the 30 benchmark cases in the final run.
- **Technical Driver of Variation:** In quantized CPU/Metal execution on Apple Silicon, latency depends directly on the number of autoregressive tokens generated in the reasoning section of the JSON payload and the thermal throttling state of unified memory.

18 B4 Compliance

Table 6 formally details compliance with the acceptance criteria established for Block B4.

Table 6: Formal compliance matrix for Block B4 acceptance criteria.

Acceptance Criterion	Target	Observed	Classification
Conflict reports with sources	100.0 %	100.0 % (12/12)	Met in development sample
False Pure Case Rate	$\leq$ 5.0 %	0.0 % (0/12)	Met in development sample
Common Trigger Identification	100.0 %	100.0 % (12/12)	Met in development sample
Colliding Intentions Identified	100.0 %	100.0 % (24/24)	Met in development sample
Explicit Source Traceability	100.0 %	100.0 %	Met in development sample
Mandatory Human Review Routing	100.0 %	100.0 % (22/22)	Met
Autonomous AI Resolution	Prohibited	Verified	Met by design and code
YAML Blocked without Approval	100.0 %	100.0 % (0 % bypass)	Met cryptographically
Dynamic BDI Cycle Execution	Operational	Pending	Not evaluated (Weeks 10–12)

19 Limitations and Overfitting Risk

In accordance with research integrity standards, the following limitations are formally declared:

- **Preliminary Nature:** Performance reflects results within the evaluated development sample and does not support immediate generalization to distinct domains or language distributions.
- **Overfitting Risk:** Because the 10 agronomic cases were consulted during hook calibration, quantitative scores do not substitute for an independent hold-out test.
- **BDI Cycle Status:** Formally maintained as:

```
runtime_compatible = "unknown"
simulation_execution_status = "not_executed"
```

  Block B4 verifies pre-simulation classification, traceability, and governance, but does not execute dynamic simulations within the BESA engine.

20 Future Work

Planned milestones for subsequent research phases include:

- **Week 8 (Block B5):** Implementation of formal eBDI deliberation for conflict resolution and alternative prioritization.
- **Independent Hold-Out Dataset:** Construction of a blind benchmark comprising at least 50 external agronomic cases.
- **Institutional Deployment:** Migration to an institutional server equipped with dedicated GPUs (24 GB VRAM) to reduce LLM latency from 55 s to $<$ 3 s per call.
- **Weeks 10–12:** End-to-end integration with the WellProdSim / EthosTerra multi-agent engine.

21 Conclusions

Week 7 successfully consolidated the Conflict-Case classifier (Block B4). The work validates a hybrid architecture wherein deterministic in-memory hooks resolve 70.0 % of requests within 5.0 ms, while the local micro-LLM provides semantic classification for ambiguous or non-trivial pure contexts.

The framework guarantees full source traceability by distinguishing contextual fields and documentary sources authenticated with SHA-256 hashes. Furthermore, the human-in-the-loop approval gate prevents stochastic autonomous resolution and cryptographically blocks unauthorized YAML generation. While development sample metrics reached 100.0 %, these results remain preliminary, motivating future blind hold-out testing and dynamic simulator integration.

22 Appendix A: Case Matrix

Table 7 presents the audited record of all 30 benchmark cases evaluated in the official run.

Table 7: Full case matrix for the Week 7 benchmark (30 cases).

Case ID	Expected	Predicted	Trigger	Decision Source	Human	Latenc
`case-001-pure`	pure	pure	N/A	DMR LLM (Gemma 4)	No	34,23
`case-002-conflict`	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
`case-003-ambiguous`	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
`case-004-pure-level`	pure	pure	N/A	DMR LLM (Gemma 4)	No	36,51
`case-005-conflict-leak`	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
`case-006-pure`	pure	pure	N/A	DMR LLM (Gemma 4)	No	69,12
`case-007-pure`	pure	pure	N/A	DMR LLM (Gemma 4)	No	68,48
`case-008-pure`	pure	pure	N/A	DMR LLM (Gemma 4)	No	53,50
`case-009-conflict`	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
`case-010-conflict`	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
`case-011-conflict`	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
`case-012-ambiguous`	unclear	unclear	Unclear	DMR LLM (Gemma 4)	Yes	50,58
`case-013-ambiguous`	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
`case-014-ambiguous`	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0

Continued on next

Table 7: Full case matrix (continued).

Case ID	Expected	Predicted	Trigger	Decision Source	Human	Latenc
case-015-hallucination	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
case-016-hallucination	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
case-017-hallucination	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
case-018-adversarial	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-019-adversarial	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-020-adversarial	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-021-agri-pure	pure	pure	N/A	DMR LLM (Gemma 4)	No	66,19
case-022-agri-pure	pure	pure	N/A	DMR LLM (Gemma 4)	No	61,13
case-023-agri-conflict	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-024-agri-conflict	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-025-agri-conflict	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-026-agri-conflict	conflict	conflict	Yes	Deterministic Hook	Yes	5.0
case-027-agri-ambiguous	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
case-028-agri-ambiguous	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
case-029-agri-hallucination	unclear	unclear	Unclear	Deterministic Hook	Yes	5.0
case-030-agri-pure	pure	pure	N/A	DMR LLM (Gemma 4)	No	55,66

23 Appendix B: Reproduction Procedure

The following conceptual commands document the verified steps required to reproduce the reported experimental results:

Listing 2: Commands for environment reproduction and evaluation suites.

```
# 1. Verify API health and DMR model runner connectivity
make health

# 2. Run canary integration test with the local micro-LLM
make canary

# 3. Execute 30-case benchmark evaluation
uv run python evals/runner.py --verbose
```

```
# 4. Execute the full 53-unit-test regression suite
PYTHONPATH=apps/api uv run --project apps/api --with pytest --with pyyaml --with
    jsonschema python -m pytest tests/unit
```

24 References

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